CRISPR-CAS9 GENE EDITING — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: April 15, 2026

Diagram-based questions on CRISPR-Cas9 Gene Editing

Question 1

10.0 points

Examine the diagram below that illustrates the Cas9 protein–sgRNA complex docking onto target DNA. Describe the molecular interactions between the sgRNA and the target DNA that are critical for specificity. Identify any key residues or structures in the Cas9 protein involved.

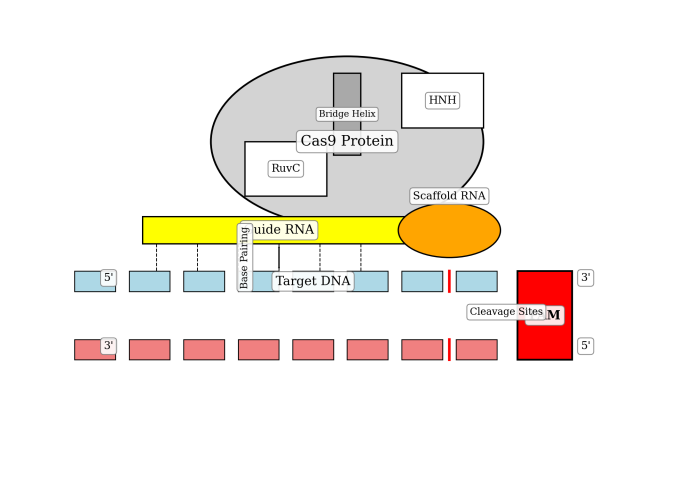

Fig. 1. Diagram of the Cas9 protein–sgRNA complex docking onto target DNA. Highlight the molecular interactions between sgRNA and target DNA that are critical for specificity. Include key residues or structures in the Cas9 protein involved in the interaction. Use labels to indicate sgRNA, target DNA, and Cas9 protein components.

Question 2

10.0 points

Examine the diagram below illustrating the PAM sequence recognition and DNA unwinding mechanism in CRISPR-Cas9. Discuss the role of the PAM sequence in Cas9 activation and how DNA unwinding is achieved.

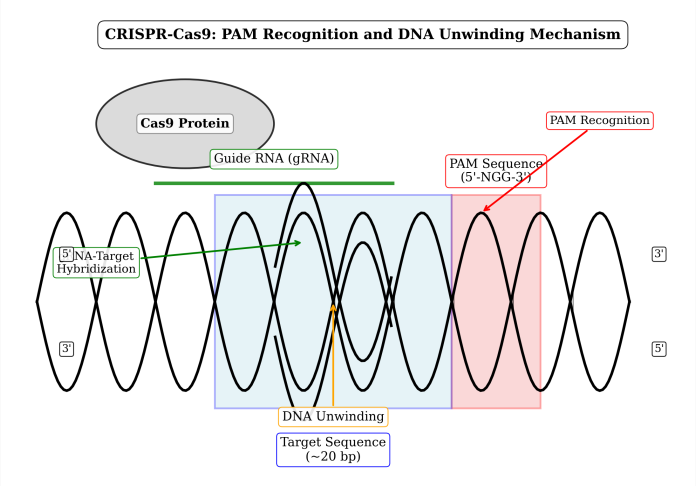

Fig. 2. A diagram illustrating the CRISPR-Cas9 mechanism. Show a DNA double helix with a highlighted PAM sequence. Include the Cas9 protein bound to the DNA, indicating the recognition of the PAM sequence. Illustrate the DNA unwinding process with arrows showing the separation of the DNA strands. Label the PAM sequence, Cas9 protein, and unwound DNA strands. The diagram should clearly depict the interaction between Cas9 and the DNA, emphasizing the role of the PAM sequence in activating Cas9 and initiating DNA unwinding.

Question 3

10.0 points

Examine the diagram below illustrating DNA double-strand break repair pathways: non-homologous end joining (NHEJ) versus homology-directed repair (HDR). Compare these pathways in terms of accuracy and potential applications in gene editing.

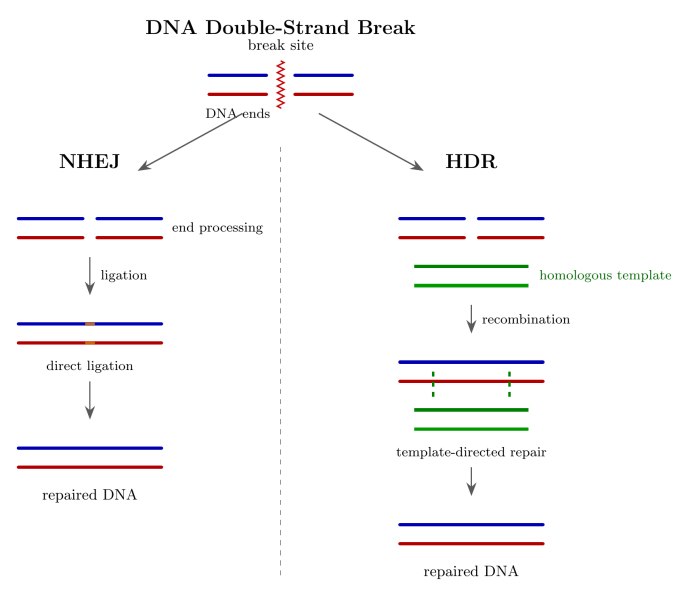

Fig. 3. Diagram illustrating DNA double-strand break repair pathways: non-homologous end joining (NHEJ) versus homology-directed repair (HDR). The diagram should show a DNA double-strand break with two separate pathways. For NHEJ, depict the direct ligation of DNA ends without a homologous template, highlighting potential for errors. For HDR, show the use of a homologous donor template for accurate repair, indicating the involvement of homologous recombination. Include labels for key steps and components in each pathway, such as DNA ends, ligation, homologous template, and recombination. Use arrows to indicate the flow of each repair process and clearly differentiate between the two pathways. IMPORTANT: The diagram is missing required labels. The diagram is missing a label for the ". Add each missing label as visible text at the appropriate location in the diagram.

Question 4

10.0 points

Analyze the mismatch tolerance in the diagram below for off-target effect prediction in CRISPR-Cas9 gene editing. How do different types of mismatches affect the likelihood of off-target effects?

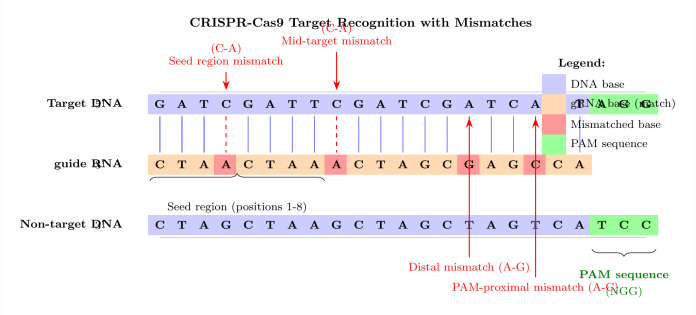

Fig. 4. A diagram illustrating the CRISPR-Cas9 complex with a guide RNA (gRNA) bound to a target DNA sequence. The diagram should show the gRNA-DNA pairing, highlighting potential mismatches. Include labels for the gRNA, target DNA, and mismatched bases. Show different types of mismatches, such as single nucleotide mismatches and bulges, to illustrate how they might affect the likelihood of off-target effects. Use arrows or annotations to indicate the mismatched regions and their potential impact on binding specificity. IMPORTANT: Values or sequences in the diagram don't match the question. The diagram contains multiple incorrect base pairings according to. Regenerate using EXACTLY the values and sequences specified. IMPORTANT: Values or sequences in the diagram don't match the question. The diagram shows multiple incorrect base pairings that violate the. Regenerate using EXACTLY the values and sequences specified.

Question 5

10.0 points

Examine the base editing window in the diagram below for cytosine base editors (CBEs). Describe the CBE's base editing window and discuss the implications for specificity and efficiency in gene editing projects.

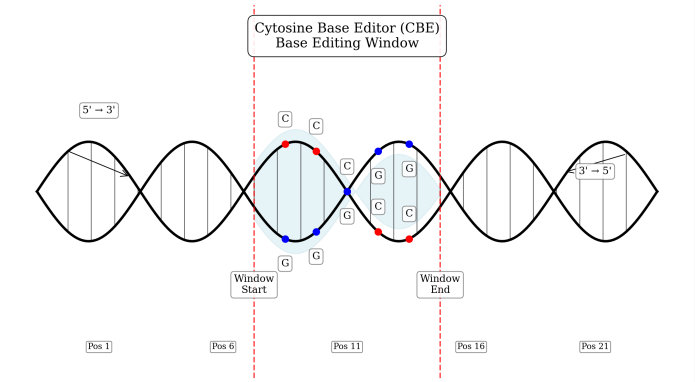

Fig. 5. Diagram showing the base editing window for cytosine base editors (CBEs) on a DNA sequence. Highlight the specific positions where cytosine bases can be edited within the sequence, illustrating the editing window's range and the implications for specificity and efficiency in gene editing projects. Include labels for the DNA sequence, cytosine positions, and the editing window boundaries. IMPORTANT: The diagram is missing required labels. The diagram is missing several labels explicitly required by the diagram. Add each missing label as visible text at the appropriate location in the diagram. IMPORTANT: The diagram is missing required labels. The diagram is missing a required label for the ". Add each missing label as visible text at the appropriate location in the diagram.